

WEEK 2 — Prototyping + Perception (Full explanation, easy words)

1) Why Prototyping exists (the “don’t waste dev time” reason)

Modern software is interactive, and whether people accept it depends heavily on the **quality of the GUI** (how easy, clear, fast, and pleasant it feels). Prototyping is used because it lets you **test and improve the UI early**, when changes are cheap.

What prototyping gives you (MEMORIZE)

Prototyping helps you:

- test & refine ideas with users before coding
 - communicate design to stakeholders (devs/clients/management)
 - find usability problems early (cheaper than fixing after launch)
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2) Sketch → Wireframe → Prototype (Sketch-to-Design Continuum)

This is the path from a messy idea in your head to something people can actually use.

A) Sketch (IDEATE)

A sketch is **rapid freehand drawing** with **no intention** of being a finished product. It supports the *process of making*, not the final design. It’s disposable.

Example: You draw 5 different home screens in 10 minutes just to explore layouts.

Memorize

- Sketch = fast, freehand, not final, used to generate many ideas.

B) Wireframe (DETAILIZE)

A wireframe is a **basic visual guide** (skeleton/blueprint) showing: structure, layout, and how pages relate. Wireframes come **before visual design** (usually no colors/typography polish).

What wireframes focus on (MEMORIZE):

- Information architecture (how info is organized)
- Structure/layout (where things are placed)
- Flow (how user moves through tasks)

C) Prototype (VISUALIZE)

A prototype is an **expression of design intent**. In digital products, it's a **simulation of final interaction** between user and interface. It can simulate one interaction or the whole app flow.

Memorize

- Prototype = simulation of interaction (not just static screens).
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3) Fidelity in prototyping (low vs high)

What “fidelity” means (MEMORIZE)

Fidelity = how closely the prototype matches the final product in realism/detail.

Fidelity varies across 3 areas (MEMORIZE this exact trio):

1. Visual design (colors, typography, spacing)
2. Content (real content vs placeholder like Lorem Ipsum)
3. Interactivity (clickable, animations, data entry)

Teams choose fidelity based on goals, completeness, and resources (time/money/skills).

A) Low-Fidelity Prototyping (Lo-Fi)

Lo-fi prototypes are often **paper-based**, use a medium unlike final (paper/cardboard/sketches/storyboard), and are **quick/cheap/easy to change**. Often users can't "really" interact; a facilitator may "act as the computer."

Storyboarding (Lo-Fi technique)

A storyboard is a **series of sketches** showing how a user progresses through a task. Often paired with scenarios; good for role-play and understanding the journey. Used early in design.

Lo-Fi Advantages (MEMORIZE)

- Quick + cheap
- Encourages iteration (easy to discard and try again)
- Focus on high-level structure/flow, not colors/fonts
- Users feel comfortable criticizing rough sketches

Lo-Fi Disadvantages (MEMORIZE)

- Some may judge it "unprofessional"
 - Less realistic user experience
 - Navigation/flow limitations; needs facilitator to "make interaction happen"
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B) High-Fidelity Prototyping (Hi-Fi)

Hi-fi prototypes are **computer-based**, look close to final, and allow realistic mouse/keyboard interactions. They represent core UI functionality; users can enter data, click, navigate like a real system.

Hi-Fi Advantages (MEMORIZE)

- More realistic interactions
- Better for collecting real performance data (time, error rate)
- User-driven (no facilitator needed)
- More persuasive for demos to stakeholders

Hi-Fi Disadvantages (MEMORIZE)

- Takes longer; needs skills/tools
 - Designers may resist change after investing effort (sunk cost)
 - Testers comment on surface details (colors) instead of structure/function
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Key conclusion (MEMORIZE the “exam style” conclusion)

- Lo-fi is best for cost + fast iteration + focusing on interaction design and information architecture.
 - Computer prototypes help detailed testing, distribution, documentation.
 - Best approach often: **combine techniques** (start lo-fi, move hi-fi).
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4) Modern prototyping landscape (tools + newer trends)

Your slides list the newer ecosystem:

Interactive prototyping tools

Figma / Adobe XD / Sketch / InVision: interactive prototypes, live collaboration, developer handoff.

AI-assisted prototyping

AI can generate wireframes/layouts from prompts, or convert sketches into editable designs (Galileo AI, Uizard).

Code-linked prototypes

Tools that can export production-ready code (Framer, FlutterFlow).

Voice + AR/VR prototyping

Voiceflow/Botmock for conversational flows; Adobe Aero/ShapesXR for spatial prototypes.

Testing/analytics + accessibility

Maze/UserTesting for remote usability; Hotjar/FullStory style analytics; accessibility tools like Stark (contrast) and Axe DevTools (screen reader checks).

5) Psychology of perception (why users “see wrong”)

Now lecture2 connects prototyping to human perception: even if your UI is “objectively there,” users may not perceive it as you expect.

“We perceive what we expect” (MEMORIZE)

Perception is biased by:

- Past experience
- Current context
- Current goals (what they’re trying to do)

A) Biased by experience (priming)

Same image can be seen as “buildings” or the word “LIFT” depending on what you were told. Once primed (e.g., “Dalmatian dog”), you can’t unsee it.

UI meaning: Users click based on habit. If your “Next” button moves, users may still *think* it’s in the old place and misclick.

B) Biased by context

A character can be read as H in “THE” and A in “CAT” because surrounding letters change interpretation.

UI meaning: Labels + nearby items change meaning. Bad grouping can mislead.

C) Biased by goals (inattentional blindness)

If user’s goal is “find the campus map,” they scan only for map-related links and may not notice a “you won \$100” banner. Goals filter perception.

Mechanisms: goals influence where we look and what features our brain becomes sensitive to.

Design implications (MEMORIZE list)

1. Avoid ambiguity (make interpretation clear; rely on conventions when needed)
2. Be consistent (same controls same place/look across screens)
3. Understand user goals (make goal-relevant info prominent at every step)

6) Gestalt principles (how users automatically “group” what they see)

Gestalt = we naturally perceive **wholes** and structure, not disconnected pieces. The principles are used to design layouts that “make sense” visually.

The 7 principles you must MEMORIZE (with meaning + UI example)

1) Proximity

Things close together feel grouped.

UI example: “Add” and “Remove” near each other; separate them from “OK/Cancel.”

2) Similarity

Things that look similar feel grouped (same color/shape/size).

UI example: all links same color so users recognize them as clickable.

3) Continuity

We prefer continuous lines/paths; we “follow” smooth forms.

UI example: IBM logo feels continuous; sliders look like one continuous range.

4) Closure

We mentally complete incomplete shapes (fill in missing parts).

UI example: stacked document icons show only edges but we see full stack.

5) Symmetry

We interpret complex scenes as simpler symmetric shapes.

UI example: symmetric shapes feel more stable/organized; can help 3D illusion.

6) Figure/Ground

We separate foreground (figure) from background (ground).

UI example: modal popup becomes figure; page dims as ground.

7) Common Fate

Things moving together feel grouped/related.

UI example: selecting multiple files and dragging them together makes them feel like one group.

Extra “teacher nitpick” point (MEMORIZE)

Gestalt principles work **together**, so a design can accidentally imply relationships you didn’t intend. A good practice is to review your UI with each principle in mind to catch unintended grouping.

WHAT TO MEMORIZE (Week 2 “Ratta List”)

If your teacher nitpicks, memorize these exactly:

1. Definitions: sketch, wireframe, prototype
 2. Fidelity definition + 3 dimensions (visual design, content, interactivity)
 3. Lo-fi vs hi-fi: characteristics + advantages + disadvantages
 4. Storyboarding definition + purpose
 5. “We perceive what we expect” + 3 bias factors (past, context, goals)
 6. Design implications list: avoid ambiguity, be consistent, understand goals
 7. Gestalt 7 principles + basic meaning + one UI example each
 8. Modern prototyping tools/trends: Figma etc, AI, code-linked, voice, AR/VR, remote testing, analytics, accessibility
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Practice Questions (with answers + clear explanations)

A) 5 MCQs

MCQ 1

A wireframe is best described as:

A) A finalized high-visual UI

- B) A basic guide showing structure/layout and relationships between pages
- C) A paper-only prototype with role-play
- D) A full working software system

Answer: B

Explanation: Wireframes are the “skeleton/blueprint,” made before visual styling.

MCQ 2

Prototype fidelity can vary by:

- A) Color only
- B) Visual design, content, and interactivity
- C) Only interactivity
- D) Only typography

Answer: B

Explanation: These are the 3 explicit dimensions of fidelity.

MCQ 3

A key advantage of low-fidelity prototypes is:

- A) They provide the most realistic performance data
- B) Users focus mainly on surface details like color
- C) They are quick/cheap and encourage iteration and honest feedback
- D) They require specialized tools and skills

Answer: C

Explanation: Lo-fi helps iterate fast and users feel comfortable criticizing.

MCQ 4

“Users don’t notice the \$100 banner because they’re searching for a campus map” demonstrates perception biased by:

- A) Symmetry
- B) Goals (inattentional blindness / filtering)
- C) Similarity
- D) Continuity

Answer: B

Explanation: Goals filter perception; unrelated info may not register.

MCQ 5

Which Gestalt principle says objects moving together are perceived as related?

- A) Closure
- B) Common Fate
- C) Figure/Ground
- D) Proximity

Answer: B

Explanation: Motion together = grouped.

B) 5 Short Answer Questions (with answers)

SAQ 1: Define prototyping in HCI. Why is it used?

Answer: Prototyping is creating a preliminary model/simulation of a product to explore ideas and evaluate quality early. It's used to test with users, communicate to stakeholders, and find usability issues cheaply before development.

SAQ 2: Explain the sketch → wireframe → prototype progression.

Answer: Sketch = fast ideation (many rough ideas). Wireframe = structured layout/flow without visual polish. Prototype = interactive simulation of final user-interface behavior.

SAQ 3: What is low-fidelity prototyping? Give one example.

Answer: Lo-fi uses a medium unlike the final product (paper/sketches/storyboards), quick and easy to change. Example: paper prototype of app screens where facilitator swaps pages when user "taps."

SAQ 4: Write the 3 factors that bias perception.

Answer: Past experience, current context, current goals.

SAQ 5: List any 4 Gestalt principles and one UI use-case for each.

Answer (sample):

- Proximity: place related buttons close.
 - Similarity: make all links same color/style.
 - Figure/Ground: modal popup with dimmed background.
 - Closure: icon stacks show partial objects but users see full objects.
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C) 1 Case Study (with model answer)

Case Study: Redesigning a university portal (prototype + Gestalt + perception)

Your university portal has: course registration, fee voucher, timetable, results. Students complain:

- "I can't find the fee voucher quickly."
- "The page looks cluttered."

Task:

1. Choose low-fidelity or high-fidelity prototyping first and justify.
2. Give a storyboard (3–4 frames described in words).

3. Identify 2 Gestalt principles you will apply to reduce clutter.
4. Use “perception biased by goals” to explain why students miss the fee voucher link.
5. Say when you would switch to high-fidelity and what you would measure.

Model Answer

1) Start with low-fidelity first

Because we need fast iteration on layout and navigation without wasting time on visual polish. Students will feel comfortable suggesting big changes.

2) Storyboard (described)

Frame 1: Student opens portal home; sees 4 main tiles (Registration, Fees, Timetable, Results).

Frame 2: Student clicks “Fees” tile; sees “Generate Voucher” and “Payment Status.”

Frame 3: Student taps “Generate Voucher”; selects semester; clicks confirm.

Frame 4: Voucher appears with download button and clear next step (“Print or pay via bank/app”).

3) Gestalt to reduce clutter

- Proximity: keep fee-related actions together; separate from unrelated items using spacing.
- Similarity: make all primary actions look consistent (same button style), so users instantly recognize what’s clickable.

4) Why students miss the voucher link (goals)

Students come with a goal (“get voucher now”), so they scan only for goal-related words. If “Voucher” is hidden under an unexpected label or placed where they don’t expect it, they won’t notice it—even if it’s on the screen.

5) When switch to high-fidelity + what measure

After agreeing on the best flow and layout, switch to hi-fi to test realistic interaction and measure task completion time and error rate (e.g., wrong clicks, backtracking).